PROJECT: QUERYQUEST

ASSIGNMENT - 1

Q1. What is the difference between a Database Management System (DBMS) and a traditional file system? Provide one concrete scenario in which a DBMS is clearly the better choice and explain your reasoning.

ANS: A DBMS System stores data in structured databases and it provides a lot of features like: querying, security and multi user access simultaneously and also maintain data consistency while a traditional file system stores data in separate files like spreadsheets with very low security and the data must be accessed manually and duplicate data may also stored in them.

For example: In a college, if a student's data is registered in both the college file and a hostel record using a traditional file system, now if he wants to update anything then any updates must be made in all files manually. However, a DBMS stores data centrally; therefore, the student can change data in just one place, and it automatically updates across all files.

Q2. In an ER diagram, what distinguishes a weak entity from a strong entity? How is the primary key of a weak entity formed and what is the identifying relationship associated with it called?

ANS: In an ER diagram a strong entity can be uniquely identified by its own attributes and represented by a single rectangle box, but the weak entity cannot be uniquely identified by its own attributes they require the attributes of strong entity, and it is represented by double rectangle.

Primary key of weak entities is formed by the combination of primary key of the strong entity(owner) and partial key of the weak entity (Dependent).

The identifying relationship associated with is represented by double diamond in between them.

Q3. A university database stores the phone numbers of students, where a student may have multiple phone numbers. Should the phone number be modeled as a simple attribute, a composite attribute or a multivalued attribute? Justify your answer.

ANS: In this, since the student have multiple phone numbers means multiple values of an attribute so phone number be modelled as a multivalued attribute.

Q4. Explain the difference between a candidate key and a super key. Consider the relation: Student (roll no, email ,name ,branch) List all candidate keys and provide at least two super keys that are not candidate keys.

ANS: Candidate keys are the minimum set of attributes required to uniquely identify each entity. Super key is also a set of attributes required to uniquely identify an entity, but it also includes some other attributes in extra that are not required for the uniqueness. So, we can say that every candidate key is super key but every super key is not a candidate key.

In the example above:

Candidate keys: {roll no} , {email};

Super keys that are not candidate key: {roll no, name} , {roll no , branch};

Q5. Suppose a row is inserted into an Enrollment table with a student_id value that does not exist in the student table.

a) Name the constraint that is violated.

b) Explain what it protects against.

c) State what should happen to the insert operation.

ANS: a) Referential Integrity constraint is violated.

b)It protects against the false entries in the Enrollment table.

c) The insertion operation will throw an error stating Referential Integrity constraint error.

Q6. When mapping a many-to-many (M: N) relationship from an ER diagram to a relational schema:

a) What additional structure must be created?

b) What does its primary key consist of?

c) Why can the relationship not simply be represented as a column in one of the participating entity tables?

ANS: a) An additional table must be created to represent the relationship between one table to another table.

b) The primary key of additional table will be the combination of the primary keys of both the tables from which we created this additional table.

c) In many to many relationships one item in a table may be connected to multiple items of another table and vice versa due to this if we want to represent the relationship as a column in one of the participating entity tables then multiple values will be stored in a single cell which violates the rule of atomicity.

PART – II

PART – A:

Entities: 1) Customer, 2) restaurant, 3) menu items, 4) orders

Key attributes: 1) Customer- customer_id

2) Restaurant – restaurant_id

3) Menu items- menuitems_id

4) Orders – order_id

Cardinality: 1) Customer-order (1:n): because a customer can give multiple orders but for a given order there can be only one customer.

2) Restaurant-order(1:n) : because a restaurant can receive multiple orders but a particular order can associated with only one restaurant.

3) Order-Menu Items(m:n): because a menu item can be included in many orders and a particular order can contain many menu items.

4) Restaurant-Menu Items(1:n): A restaurant can have multiple menu items but many items can be associated with only one restaurant.

PART B:

Multivalued attribute: Phone Number (a person can have more than one phone number)

Composite attribute: Address (an address may also have multiple attributes in it like house_no , street_no etc.)

PART C:

For the junction table the primary key will be the combination of primary keys of both the entities;

primary key = (order_id, menu_items_id)

foreign keys = order_id and menu_items_id

additional attribute: Quantity

Order_menuItems (order_id , menu_items_id , Quantity)

PART – III

Q-2:

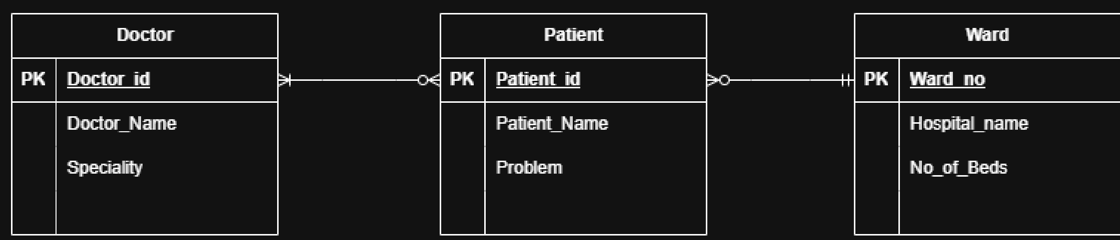
ANS: In this the Doctor Patient relationship is M:N because a Doctor can treat multiple patients and a patient can be treated by multiple doctors, they create a junction table in which the primary key would be the combination of both the Doctor_id and the patient_id(primary key of both the individual entities).

Patient-Doctor(M:N): In this the patient's participation is full a patient must be assigned to at least one doctor but the doctor participation is partial because it may be the condition where a doctor does not have any patient to treat.

Patient-Ward(N:1): The patient participation is full because a patient will give exactly one ward, but the ward participation is partial because a ward may not have any patient at any time.

An address attribute was made composite it also has a lot of other attributes associated with it like (city,road_no,street etc) and a phone no is multivalued because a person may have more than one phone number.

Q-1:


Doctor
PK
Doctor_id
Doctor_Name
Speciality
Patient
PK
Patient_id
Patient_Name
Problem
Ward
PK
Ward_no
Hospital_name
No_of_Beds